1.Create different blinking patterns with LEDs by connecting to GPIO pints of Raspberry Pi

Aim: Create different blinking patterns with LEDs by connecting to GPIO pints of Rasberry Pi.

Procedure: install Python3 on Raspi.

Command:

sudo apt install idle3

ii) Connection of LED to Raspi:

Connect LEDs on breadboard.

Connect resistor to one end of LED (to Positive end or anode).

Connect other end of resistor to GPIO pin on Raspberry Pi.

Connect other end of LED to ground (negative end or cathode).

Code;

```
import RPi.GPIO as GPIO
import time

GPIO.setmode(GPIO.BOARD)
GPIO.setup(3, GPIO.OUT)

for i in range(10):
  GPIO.output(3, True)
  print("led on")
  time.sleep(0.5)
  GPIO.output(3, False)
  print("led off")
  time.sleep(0.5)

print("complete")
GPIO.cleanup()
```

Interface Pi Camera with Raspberry Pi to capture image and video

Aim; Interface Pi Camera with Raspberry Pi to capture image and video

Procedure ;

Connect the finger print module to any one of the USB port of Raspi.

i) Install Python3 on Raspi.

command prompt:

sudo apt install idle3

sudo apt install python3-picamera2 -y

sudo apt install libcamera-apps -y

pip install opencv-python

pip install numpy

code;

image code

```
from picamera2 import Picamera2
import time

picam2 = Picamera2()
picam2.start()
time.sleep(2)
picam2.capture_file("photo1.jpg")
print("Image Captured Successfully")
```

video code

```
from picamera2 import Picamera2
import time

picam2 = Picamera2()
picam2.start_and_record_video("video.mp4")
time.sleep(10)
picam2.stop_recording()
print("Video Saved")
```

Interface 16x2 LCD with Raspberry Pi using I2C module to display text

Procedure;

i) Enable I2C Interface on Raspberry Pi by writing below code in command terminal

sudo raspi-config

ii) Navigate:

Interface Options

→ I2C

→ Enable

iii) Reboot Raspi by writing below code in command terminal

sudo reboot

iv) Install I2C Tools by writing below code in command terminal

sudo apt update

sudo apt install -y i2c-tools python3-smbus

Verify whether LCD is detected

sudo i2cdetect -y 1

v) Install LCD Library on Raspberry Pi by writing below code in command terminal

pip3 install RPLCD

Code;

```
from RPLCD import CharLCD
from RPi import GPIO
from time import sleep

lcd = CharLCD(
    numbering_mode=GPIO.BCM,
    cols=16,
    rows=2,
```

```
    pin_rs=26,
    pin_e=19,
    pins_data=[13, 6, 5, 11]
)

lcd.clear()
lcd.write_string("HELLO")
sleep(2)
lcd.clear()
lcd.write_string("Welcome")
lcd.cursor_pos = (1, 0)
lcd.write_string("PRASAD")
sleep(5)
lcd.clear()
```

Interface ADS 1115 with Raspberry Pi to act as oscilloscope.

Procedure;

Step 1: Enable Raspberry Pi I2C interface

To enable the I2C, from the terminal, run:

sudo raspi-config

When the configuration panels open, select Interface Options, select I2C and click Enable.

Step 2: Update the Raspberry Pi

To do this, run below two commands:

sudo apt update

sudo apt upgrade -y

Step 3: Install the Adafruit ADS1115 library for ADC and Matplotlib for plotting of graph.

pip3 install adafruit-circuitpython-ads1x15

pip3 install matplotlib

Code;

```
import time
import board
import busio
import adafruit_ads1x15.ads1115 as ADS
from adafruit_ads1x15.analog_in import AnalogIn
# Create I2C bus
i2c = busio.I2C(board.SCL, board.SDA)
# Create ADS1115 object
ads = ADS.ADS1115(i2c)
# Read Analog Channel A0
chan = AnalogIn(ads, 0)
print("Digital oscilloscope Started")
print("---------------------------")
while True:
    print("Voltage = %.3f V" % chan.voltage)
    time.sleep(0.2)
```

Interface TM1637 with Raspberry Pi and Display current time

Procedure;

Install libraries of TM1637 as tm1637.py which is driver library. Download and save it in same folder as the main program code.

Code;

```
import time

import datetime

import RPi.GPIO as GPIO

import tm1637

CLK = 23

DIO = 24

display = tm1637.TM1637(CLK, DIO, brightness=7)

while True:

    now = datetime.datetime.now()

    hour = now.hour

    minute = now.minute

    second = now.second

    display.numbers(hour, minute)

    time.sleep(1)


##CLK → GPIO 23 (physical pin 16)

##DIO → GPIO 24 (physical pin 18)

##VCC → 3.3V or the voltage supported by your module

##GND → GND
```

Interface GPS module with Raspberry Pi.

Procedure;

Procedure: Interface GPS with Raspberry Pi

1. Enable UART by entering:
sudo raspi-config

2. Go to Interface Options → Serial Port. Select No for login shell and Yes for serial hardware.

3. Reboot Raspberry Pi:
sudo reboot

4. Verify the serial port:
ls /dev/serial*

5. Install GPS packages:
sudo apt update

sudo apt install gpsd gpsd-clients python3-gps -y

6. Stop and disable GPS daemon:
sudo systemctl stop gpsd.socket

sudo systemctl disable gpsd.socket

7. Start GPS daemon:
sudo gpsd /dev/serial0 -F /var/run/gpsd.sock

8. Test GPS data:
cgps -s

9. Write and run the Python program gps2.py to read GPS data.

Code;

```
import serial

import pynmea2


ser = serial.Serial("/dev/serial0", 9600, timeout=1)

print("Waiting for GPS signal...\n")
```

```
try:
    while True:
        data = ser.readline().decode("ascii", errors="ignore").strip()
        if data.startswith(("$GPGGA", "$GNGGA")):
            try:
                msg = pynmea2.parse(data)
                if msg.gps_qual > 0:
                    print("GPS Fix Available")
                    print("Latitude :", msg.latitude)
                    print("Longitude:", msg.longitude)
                    print("Altitude :", msg.altitude, "meters")
                    print("-----------------------")
                else:
                    print("Waiting for satellite fix...")
            except pynmea2.ParseError:
                pass
except KeyboardInterrupt:
    print("\nProgram Stopped")
finally:
```

Use Telegram to control Raspberry Pi GPIO

Steps;

Download Telegram from playstore on your android phone.

i) Install Telegram.

ii) Open Telegram app in your system or mobile

iii) Click On Start Messaging Button

iv) Enter your mobile number to register with telegram service.

v) Search for name "BotFather".

Procedure;

Set up On Raspberry Pi:

1. Install "Python Package Index" and Telepot using :

python3 -m pip config set global.break-system-packages true

sudo apt install python3-pip

pip3 install telepot

2. Test your bot using python3 IDLE and type :

import telepot

bot = telepot.Bot('Bot Token')

bot.getMe()

code;

```
import RPi.GPIO as GPIO
import telepot
from telepot.loop import MessageLoop
import time
TOKEN = "8781695804:AAFgeuxHZtSx70EPq0GrmR_ML5xJzG8Yc2g"
GPIO.setmode(GPIO.BCM)
LED = 2
GPIO.setup(LED, GPIO.OUT)

# Start with LED OFF
GPIO.output(LED, GPIO.LOW)
```

```
bot = telepot.Bot(TOKEN)

def handle(msg):
    chat_id = msg["chat"]["id"]
    text = msg.get("text", "").lower()

    if text == "/on":
        GPIO.output(LED, GPIO.HIGH)
        bot.sendMessage(chat_id, "LED is ON")

    elif text == "/off":
        GPIO.output(LED, GPIO.LOW)
        bot.sendMessage(chat_id, "LED is OFF")

    else:
        bot.sendMessage(chat_id, "Please send /on or /off")

MessageLoop(bot, handle).run_as_thread()

print("Bot is running...")
print("Send /on or /off to your Telegram bot.")

try:
    while True:
        time.sleep(1)
except KeyboardInterrupt:
    print("\nBot stopped.")
finally:
```

GPIO.cleanup()

Interface Fingerprint Sensor and 16X2 LCD with Raspberry and display message RFID UID matched or unmatched.

Procedure;

Connect the finger print module to any one of the USB port of Raspi.

[Image: Fingerprint module connected to Raspberry Pi USB]

i) Install Python3 on Raspi.

Write the below code in command prompt :

```
sudo apt install idle3
```

Setting up Raspberry Pi for Serial Communication

```
sudo raspi-config
```

In the "Interfacing Options", select the "UART" option.

Select "Yes" option.

Finish the process and Reboot the Raspberry Pi.

Following are the terminal command that needs to be executed.

```
sudo apt update
pip install pyfingerprint
```

code

```
from pyfingerprint.pyfingerprint import PyFingerprint
import time


try:
    try:
```

```
    f = PyFingerprint('/dev/ttyUSB0', 57600, 0xFFFFFFFF, 0x00000000)
except:
    f = PyFingerprint('/dev/serial0', 57600, 0xFFFFFFFF, 0x00000000)

if not f.verifyPassword():
    raise Exception("Wrong sensor password")

print("Sensor connected!")
print("Stored fingerprints:", f.getTemplateCount())

print("Place finger...")
while not f.readImage():
    time.sleep(0.1)
f.convertImage(1)

print("Remove finger...")
while f.readImage():
    time.sleep(0.1)

print("Place same finger again...")
while not f.readImage():
    time.sleep(0.1)
f.convertImage(2)

if not f.compareCharacteristics():
    raise Exception("Fingerprints do not match")

f.createTemplate()
```

```
    pos = f.storeTemplate()
    print("Fingerprint stored at position:", pos)

except Exception as e:
    print("Error:", e)
```